

Block-Diffusion OD Planning — v4 Results (EOS-fixed backbones)

2026-07-27 · This report is built **entirely on the v4 round**: all seven mask backbones retrained after the EOS boundary fix, all discriminators and evaluations redone on top of them. The previous consolidated report (RESULTS_BD.pdf) remains the record of the pre-fix (v2) backbones and is referenced here only as the before-side of the fix. **Naming**: every v4 artifact carries the `_v4` tag — checkpoints `BD_porto_v3_normal_mask_blk{B}_v4_bd.pth` (the `v3` token is the *dataset* index `porto v3-0.05`, not a model round), discs `BDdisc_*_v4.pth`, pools `*_pool_v4_*`, eval tags `v4m* / v4f01m*`.

1. The EOS boundary fix

The training canvas is `[dst, v0, ..., vL-1=dst, END...END | PAD...]`, where `END` fills only the remainder of the block containing `dst` and later blocks are `PAD` (excluded from loss). Under that spec a sample whose `dst` lands **exactly on a block boundary** carries **zero END supervision** — a 1/B fraction of the data, and for B=1 every sample. The v2 blk1 backbone therefore never learned to terminate: unconditionally, 100% of its generations ran to the canvas cap (length 101, 50.5% invalid edges, valid rate 0.000). At B≥2 the same defect appeared as a systematic deficit of boundary-ending lengths: the generated share of lengths with $(L+1) \bmod B = 0$ was roughly half the real share (blk2 0.234 vs 0.500; blk32 0.011 vs 0.027).

`Fix ( bd_models.py build_canvas() )`: when the `END` tail would be empty, **extend the canvas by one full block of END tokens, included in the loss**. Every sample now trains ≥1 `END` target; for B=1 that is exactly one `END` token per sample. Verified by `smoke_test_bd.py` (canvas spec, both kernels' forward/backward and planning).

blk64 is unchanged by construction: no real path in this dataset has L=63 or 127 (boundary share ≈ 0.000), so the fix touches zero blk64 training samples; with identical seeding the v4 blk64 backbone reproduces v2's numbers exactly. It is retained for completeness.

2. Unconditional generation — before / after

20,000 open-ended generations per backbone (`plan(None, None)`), valid under 10% joint ori+dst condition dropout), scored on the `NORMAL` world. Real reference: length 23.2±9.8. **valid** = all edges legal; **invE** = invalid-edge rate; **KS** = length-distribution Kolmogorov-Smirnov distance vs real; **cap%** = generations hitting the canvas cap (failed to stop); **novel** = share not appearing verbatim in the real set; **bres** = boundary-residue share, generated/real — the length class starved of `END` supervision before the fix. v2 values in gray.

blk	valid v4 (v2)	invE v4 (v2)	len v4 (v2)	KS v4 (v2)	cap% v4 (v2)	novel v4 (v2)	bres gen/real v4 (v2 gen)
1	0.870 (0.000)	2.29% (50.47%)	25.2 (101.0)	0.045 (1.000)	0.9 (100.0)	0.396 (1.000)	—
2	0.737 (0.665)	3.30% (4.41%)	23.3 (24.1)	0.010 (0.042)	0.0 (0.1)	0.427 (0.519)	0.486/0.500 (0.234)
4	0.617 (0.517)	2.59% (3.78%)	23.8 (24.1)	0.026 (0.039)	0.0 (0.1)	0.662 (0.747)	0.235/0.250 (0.173)
8	0.325 (0.382)	5.88% (4.86%)	23.8 (24.5)	0.023 (0.055)	0.0 (0.0)	0.855 (0.834)	0.115/0.125 (0.102)
16	0.202 (0.184)	8.30% (9.09%)	24.4 (23.0)	0.056 (0.019)	0.0 (0.0)	0.902 (0.903)	0.054/0.061 (0.033)
32	0.141 (0.232)	9.37% (11.48%)	25.6 (16.7)	0.120 (0.278)	0.0 (0.0)	0.945 (0.935)	0.032/0.027 (0.011)
64	0.104 (0.104)	11.42% (11.42%)	24.1 (24.1)	0.056 (0.056)	0.0 (0.0)	0.971 (0.971)	0.000/0.000 (0.000)

- blk1 collapse fully resolved**: valid 0.000 → 0.870 (now the best unconditional generator), invalid edges 50.5% → 2.3%, length 101 (all capped) → 25.2 with KS 0.045. AR needed exactly one trained `END` per sample.
- Boundary-residue deficit gone** at every block: generated shares now match real (blk2 0.486 vs 0.500; before 0.234).
- Validity–novelty trade-off persists**: small blocks are more valid but copy training routes more often (blk1 novel 0.396); blk4–8 balance validity 0.33–0.62 with novelty 0.66–0.86.
- Caveats**: blk8 (0.382→0.325) and blk32 (0.232→0.141) validity dipped — single-run training variance is not separable here; blk32's length calibration nevertheless improved sharply (KS 0.278→0.120, mean 16.7→25.6).

2b. Unconditional EM/PC — each sample's own endpoints as its OD

EM/PC need no *given* OD: `evaluate_em_pc` takes the generated path's own (start, end) as its OD and looks it up in the reference shortest-path data. Since training paths are shortest paths, **uncond EM** = "is the sample a shortest path between its own endpoints" — a distribution-fidelity metric. OD coverage (share of generated ODs present in the reference) is ≈1.0 at every block, so the measurement is clean. 3,000 samples/block, seed 7. **novel^EM** is the synthetic-data currency: a *new* shortest path for a known OD, i.e. an alternative equal-length route absent from the dataset.

blk	valid	EM	PC	novel	valid^novel	novel^EM	OD cov
1	0.864	0.640	0.716	0.406	0.271	0.046	1.000
2	0.745	0.629	0.675	0.425	0.170	0.054	1.000
4	0.633	0.401	0.481	0.652	0.285	0.053	0.999
8	0.346	0.178	0.230	0.851	0.197	0.029	0.998

16	0.197	0.113	0.141	0.907	0.104	0.019	0.997
32	0.130	0.058	0.080	0.951	0.081	0.009	0.998
64	0.105	0.035	0.056	0.973	0.078	0.008	0.995

- **blk1/2 dominate shortest-path fidelity** (EM 0.63–0.64 vs blk4 0.40); on the novelty-quality frontier blk2/4 hold only a slim edge (novel $\wedge$ EM 0.053–0.054 vs blk1's 0.046; valid $\wedge$ novel blk4 0.285 vs blk1 0.271).
- **Novelty caveat:** this dataset is shortest paths, so per OD the answer is near-unique — a *perfect* generative model would show *low* novelty. blk1's verbatim copies are partly a sign of learning the distribution, not only memorization; the novelty axis is intrinsically tilted toward larger blocks, and even so their edge is $\leq 1\%$ absolute.

2c. Diversity among VALID generations — the intuition test

Raw novelty rises with block size (0.406 $\rightarrow$ 0.973, within-block mean-field sampling adds combinatorial variety) — but diversity of invalid paths is noise. Restricting to the valid subset and benchmarking against an equal-size real sample (metrics: distinct-OD share, distinct directed edges used, edge-usage Shannon entropy, mean pairwise Jaccard similarity of edge sets — higher = more mutually similar):

	n valid	distinct OD	edge support	edge entropy	pairwise Jaccard
real (20k)	20,000	1.000	3,856	10.62 b	0.0124
blk1	17,395	0.995	3,793	10.54 b	0.0132
blk2	14,744	0.996	3,728	10.47 b	0.0151
blk4	12,338	0.995	3,753	10.49 b	0.0131
blk8	6,501	0.996	3,541	10.30 b	0.0149
blk16	4,046	0.995	3,273	10.09 b	0.0189
blk32	2,829	0.995	3,124	9.99 b	0.0186
blk64	2,080	0.998	2,889	10.03 b	0.0186

- **Conditioned on validity, diversity decreases with block size** — the opposite of the raw-novelty picture: edge support 3,793 $\rightarrow$ 2,889, entropy 10.54 $\rightarrow$ 10.03 b, pairwise similarity 0.0132 $\rightarrow$ 0.0186. Selection effect: at large B only 10–14% of samples are valid, and the survivors concentrate on short, common corridors.
- **blk1's valid set matches the real distribution's diversity almost exactly** (98% of the real edge support, entropy -0.08 b, near-identical pairwise similarity). The apparent diversity of large blocks is mostly the diversity of invalid noise.
- **Net reading for the block-size choice:** quality, fidelity and *valid* diversity all favor small blocks; the only quantitative diffusion edge left in open-ended generation is the slim novel $\wedge$ EM margin of blk2–4. (Sampling-speed differences measured earlier are implementation overhead, not algorithmic — first-hitting NFE is block-size-independent; see §7.)

3. Conditional planning — held-out real OD pairs (normal world)

1,000 held-out real shortest-path pairs, 3 generation seeds, mean $\pm$ std (`eval_bd_sweep_controlled -suffix v4`). **arr** = reaches destination; **v $\wedge$ h** = valid $\wedge$ hit; EM/PC per §1.2 of the v3 report. All RAW (no refine).

blk	arrival	valid	valid $\wedge$ hit	EM	PC	s/path
1	0.994 $\pm$ 0.002	0.952 $\pm$ 0.005	0.948 $\pm$ 0.005	0.843 $\pm$ 0.007	0.887 $\pm$ 0.006	0.082
2	0.948 $\pm$ 0.005	0.818 $\pm$ 0.006	0.789 $\pm$ 0.008	0.756 $\pm$ 0.005	0.782 $\pm$ 0.006	0.060
4	0.937 $\pm$ 0.004	0.878 $\pm$ 0.008	0.826 $\pm$ 0.005	0.786 $\pm$ 0.010	0.823 $\pm$ 0.010	0.035
8	0.932 $\pm$ 0.012	0.728 $\pm$ 0.003	0.687 $\pm$ 0.010	0.671 $\pm$ 0.004	0.695 $\pm$ 0.003	0.030
16	0.926 $\pm$ 0.000	0.676 $\pm$ 0.013	0.641 $\pm$ 0.008	0.600 $\pm$ 0.010	0.631 $\pm$ 0.011	0.031
32	0.879 $\pm$ 0.004	0.585 $\pm$ 0.010	0.529 $\pm$ 0.011	0.499 $\pm$ 0.006	0.533 $\pm$ 0.008	0.032
64	0.839 $\pm$ 0.005	0.430 $\pm$ 0.014	0.373 $\pm$ 0.010	0.366 $\pm$ 0.012	0.391 $\pm$ 0.012	0.033

Conditional quality still favors small blocks (dst-anchored first-hit termination is untouched by the EOS fix). Together with §2, the division of roles is now clean: **conditional planning** $\rightarrow$ **small blocks; unconditional / open-ended synthesis** $\rightarrow$ **diffusion blocks only** (blk1 was unusable there before the fix and is competitive after it, but larger blocks retain the novelty edge and guidance headroom).

4. Guidance on the exceptional scenario — fam 0.05 (except_0, RAW)

Reserved 1,000 except_0 pairs (never seen by any disc), seed 7, first-hitting, per-block model-negative discriminators (`_v4`), n_is=100. Columns per method: valid / arrival / **valid $\wedge$ arrival** / EM / PC, all RAW. Plain IW is omitted from the tables (base < IW < adj+IW ordering holds at every block; e.g. blk4 valid 0.279/0.332/0.389). **Seen** = disc trained on except_0 (1%, excluding the reserved rows); **Unseen** = disc trained on except 1–99, applied zero-shot.

4.1 Seen

blk	base valid	base arr	base v∧a	base EM	base PC	adj+IW valid	adj+IW arr	adj+IW v∧a	adj+IW EM	adj+IW PC
1	0.302	0.994	0.300	0.271	0.283	0.928	0.567	0.540	0.374	0.425
2	0.251	0.949	0.238	0.233	0.240	0.492	0.846	0.401	0.388	0.423
4	0.279	0.932	0.262	0.243	0.258	0.389	0.873	0.328	0.334	0.357
8	0.229	0.938	0.214	0.202	0.214	0.331	0.862	0.273	0.295	0.309
16	0.206	0.923	0.194	0.180	0.190	0.298	0.865	0.250	0.230	0.255
32	0.185	0.870	0.160	0.157	0.168	0.265	0.825	0.215	0.219	0.237
64	0.141	0.827	0.119	0.129	0.134	0.229	0.795	0.162	0.176	0.198

4.2 Unseen (disc = except 1–99 → except_0; base identical to §4.1)

blk	base valid	base arr	base v∧a	base EM	base PC	adj+IW valid	adj+IW arr	adj+IW v∧a	adj+IW EM	adj+IW PC
1	0.302	0.994	0.300	0.271	0.283	0.925	0.558	0.530	0.362	0.412
2	0.251	0.949	0.238	0.233	0.240	0.488	0.848	0.415	0.367	0.408
4	0.279	0.932	0.262	0.243	0.258	0.390	0.875	0.331	0.335	0.357
8	0.229	0.938	0.214	0.202	0.214	0.331	0.863	0.273	0.295	0.308
16	0.206	0.923	0.194	0.180	0.190	0.300	0.868	0.254	0.237	0.261
32	0.185	0.870	0.160	0.157	0.168	0.268	0.828	0.219	0.222	0.240
64	0.141	0.827	0.119	0.129	0.134	0.224	0.798	0.162	0.178	0.197

- The v2 conclusions replicate on the fixed backbones: adj+IW improves the strict raw success (valid∧arrival) at every block, trading arrival for a larger validity gain; blk1 is the extreme case (valid 0.302→0.928, arrival 0.994→0.567, v∧a 0.300→0.540).
- Seen/unseen gap ≤0.015 everywhere (e.g. v∧a blk1 0.540/0.530, blk4 0.328/0.331) — zero-shot scenario transfer confirmed again; the discriminator-level analysis is in DISC_VALIDITY.pdf .
- adj+IW ESS ≈ 95.5 (blk64) – 99.7 (blk1): the disc tilt stays gentle per reveal; the blk1 jump is driven by Lemma-3 adjacency masking, not the disc.

5. Guidance — fam 0.1 (except_0, RAW; same protocol)

Same pipeline at edge-remove ratio 0.1: a much harder scoring world (twice the removed edges) — base validity drops to 0.04–0.12. Per-block _v4 discs trained on fam 0.1 scenarios; unseen = except 1–99 → except_0 zero-shot.

5.1 Seen

blk	base valid	base arr	base v∧a	base EM	base PC	adj+IW valid	adj+IW arr	adj+IW v∧a	adj+IW EM	adj+IW PC
1	0.115	0.993	0.114	0.108	0.110	0.927	0.250	0.234	0.144	0.177
2	0.096	0.960	0.092	0.089	0.092	0.245	0.717	0.163	0.174	0.200
4	0.089	0.949	0.084	0.075	0.081	0.169	0.841	0.137	0.141	0.152
8	0.077	0.933	0.071	0.073	0.074	0.143	0.804	0.101	0.123	0.131
16	0.075	0.934	0.069	0.067	0.071	0.132	0.819	0.091	0.103	0.115
32	0.072	0.884	0.061	0.062	0.066	0.118	0.793	0.076	0.088	0.099
64	0.040	0.865	0.032	0.034	0.036	0.119	0.744	0.068	0.094	0.105

5.2 Unseen (base identical to §5.1)

blk	base valid	base arr	base v∧a	base EM	base PC	adj+IW valid	adj+IW arr	adj+IW v∧a	adj+IW EM	adj+IW PC
1	0.115	0.993	0.114	0.108	0.110	0.932	0.248	0.232	0.145	0.177
2	0.096	0.960	0.092	0.089	0.092	0.243	0.720	0.164	0.174	0.198
4	0.089	0.949	0.084	0.075	0.081	0.168	0.841	0.136	0.141	0.151
8	0.077	0.933	0.071	0.073	0.074	0.145	0.803	0.101	0.124	0.132
16	0.075	0.934	0.069	0.067	0.071	0.134	0.819	0.092	0.102	0.115

32	0.072	0.884	0.061	0.062	0.066	0.117	0.800	0.079	0.084	0.097
64	0.040	0.865	0.032	0.034	0.036	0.121	0.745	0.070	0.095	0.105

- **Guidance survives the harder world:** relative gains grow (blk64 valid 0.040→0.119, ~3×; v_Λa 0.032→0.068–0.070), though absolute levels are low — at 10% edge removal the base model's prior is badly mismatched with the scenario graph.
- **The arrival cost of hard masking grows with difficulty:** blk1 adj+IW arrival collapses to 0.25 (legal-only reveals overrun the destination far more often when twice the edges are gone). v_Λa still doubles over base (0.114→0.234). Endpoint post-processing (P3) is the natural complement, as in the v3 report.
- **Zero-shot transfer holds at fam 0.1 too:** seen/unseen gap ≤0.003 in v_Λa across all blocks.
- adj+IW ESS 97.2 (blk64) – 99.6 (blk1); orderings base < IW < adj+IW hold at every block.

6. Discriminator validity (summary)

Full analysis in `DISC_VALIDITY.pdf` / `DISC_VALIDITY_K0.pdf`. Headlines: (i) seen vs unseen discs score held-out inputs near-identically (logit Pearson 0.81–0.98); (ii) on the model-negative task, zero-shot AUC loss is ≤0.05 in both probe directions — most of the signal (edge legality vs the input adjacency, model artifacts) is scenario-agnostic by construction; (iii) the harder data-negative probe (real vs real) separates the two training regimes: the single-scenario e0 disc drops –0.100 (fam 0.05) / –0.084 (fam 0.1) AUC zero-shot, while the 99-scenario e99 disc is nearly flat — multi-scenario training is what buys genuine scenario-specific generalization; (iv) the v2 blk1 "best AUC" was an artifact of its degenerate negative pool and disappears after the EOS fix (0.985 → 0.870), confirming the negative-pool confound analysis.

7. Notes & provenance

- Pipeline: `run_v4_all.sh`, single tmux chain, 2026-07-27 00:48–04:22 KST (Stage A retrain ≈ 1h40m; 28 guided evals ≈ 1h). Raw artifacts: `sets_res/va_table_v4_f005.json`, `va_table_v4_f01.json`, `uncond_pool_eval_v4.json`, `disc_gen_eval_f{005,01}{,_data}_v4.json`, logs `sets_log/v4_*.log`.
- Training identical to v2 except the canvas fix: `train_porto_bd.sh mask B gpu v4` (1 epoch, bs 32, lr 5e-4, drop_cond 0.1 — ori and dst are always dropped *jointly*, never individually).
- The conditional-eval script's output prefix is hard-coded `V2_`; its per-path CSVs under that prefix now hold v4 results (v2 aggregates preserved in `sets_log/v2_ctrl_mask.log` and the v3 report).
- **s/path caveat:** with first-hitting the algorithmic NFE is block-size-independent (one forward per reveal, total ≈ generated length). The measured blk1 slowdown (0.082 vs ~0.033 s/path) is per-block implementation overhead — a CPU sync + Python stop-scan + attention-mask rebuild per block, paid per *token* at B=1 — plus batch-straggler synchronization; not an algorithmic advantage of larger blocks (no KV caching / parallel within-block commits are implemented).
- Old (pre-fix) checkpoints/discs/pools were left in place; nothing v4 overwrites them (distinct names). They can be deleted once this report is accepted.